

3.5 Scatter Plots and Linear Relationships

Lesson Introduction

 Benchmark	MA.912.DP.2.4 Fit a linear function to bivariate numerical data that suggests a linear association and interpret the slope and y -intercept of the model. Use the model to solve real-world problems in terms of the context of the data.		 Learning Targets	- I can informally fit a straight line to model the relationship between two variables in a scatter plot. - I can interpret the slope and y-intercept of a linear function fit to bivariate numerical data in context. - I can determine an appropriate domain and range for bivariate numerical data based on the context.	
 Benchmark Coherence	Building On	Working Toward			
	MA.8.DP.1.2 Given a scatter plot within a real-world context, describe patterns of association.	MA.912.DP.2.8 Fit a quadratic function to bivariate numerical data that suggests a quadratic association, and interpret any intercepts or the vertex of the model. Use the model to solve real-world problems in terms of the context of the data. MA.912.DP.2.9 Fit an exponential function to bivariate numerical data that suggests an exponential association. Use the model to solve real-world problems in terms of the context of the data.			
 Essential Questions	- How can you informally fit a straight line to model the relationship between two variables in a scatter plot? - How do you interpret the slope and y-intercept of a linear function fit to bivariate numerical data in context? - How do you determine an appropriate domain and range for bivariate numerical data based on the context?				
 Lesson Narrative	In this lesson, students review from prior courses how to informally fit a straight line to model the relationship between two variables in a scatter plot, and how to describe patterns of association. Students will begin with a refresher on scatter plots and their data. They will then collaborate to determine and describe the patterns of association that exist for data represented on given scatter plots. The lesson concludes by formally introducing students to interpreting the slope and the y -intercept of the line of best fit in context.				
 Component Summary	3.5.1 Warm-Up (~5 min.)	Find unit rate and relate it to the slope of a line (<i>SE p. 132</i>)	3.5.3 Collaboration (10-15 min.)	Describe patterns of association in data represented on scatter plots (<i>SE p. 133</i>)	
	3.5.2 Refresher (10 min.)	Review representing bivariate numerical data on scatter plots (<i>SE p. 132</i>)	3.5.4 Guided Instruction (10-15 min.)	Introduction to interpreting the slope and y -intercept of the line of fit in the context of data (<i>SE p. 136</i>)	
	3.5.5 Wrap-Up (~5 min.)	Make an argument for which scatter plot does not belong (<i>SE p. 137</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Bivariate data - Scatter Plot - Line of Fit - Association <u>Additional Terms:</u> N/A		(Optional) Graphing utility		(Optional) Anchor chart for question 3 in the 3.5.3 Collaboration

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may choose a line that is not the line of fit. - Students may reverse positive and negative association in data.	- Students learned to fit lines to data in the prior course (Grade 7 Accelerated Math or Grade 8 Pre-Algebra). - For pacing purposes, consider an increase in student discussions in the first two activities, as students have done this initial work in a prior grade. - Consider deliberate connections between what students learned in the first four lessons about linear equations and how they apply when modeling relationships between bivariate data. For example, there are opportunities to make connections between slope and the direction of association as well as the form of the line that equations in this lesson are given in.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Which One Doesn't Belong? 3.5.5 Wrap-Up provides students with four different scatter plots, and they are asked to determine which of the four does not belong. Students are prompted to explain their rationale for deciding that more than one option doesn't belong using the key terms reviewed in the lesson.	MA.K12.MTR.6.1: Reasonableness In this lesson, students informally determine the line of fit for given data sets and use them to describe patterns of association. Multiple lines are given, and students must visually assess the reasonableness of the fit of a line. They can evaluate their results based on the given context and determine if the line chosen makes sense based on the data.
 Differentiate Instruction	Consider using a graphing utility to show students how different lines can be drawn, but how the line of best fit is the line that best represents the data. Use this tool to “turn on” lines so students can see the difference between the way the data looks on different lines compared to the line of fit.	
Lesson Notes:		

3.5.1 Warm-Up: Bell Work

Component Overview: Students will complete a table with four, real-world scenarios to determine the unit rate and the slope of the line that represents the context.



3.5.1 Warm-Up: Bell Work

- Complete the table by calculating the value that completes each statement given the context and identifying the slope of the line that represents the context.

Scenario	Unit Rate Statement	Slope of the Line Representing the Context
Five granola bars cost \$20.	A granola bar costs \$ <u>4</u> .	4
A car travels 100 miles at a constant speed for 2.5 hours.	The car is traveling <u>40</u> miles per hour.	40
Tyler can do 50 sit-ups in 4 minutes.	On average, Tyler does <u>12.5</u> sit-ups per minute.	12.5
3 ounces of yeast flakes cost \$4.29.	One ounce of yeast flakes costs \$ <u>1.43</u> .	1.43



Consider adding an additional Notice and Wonder component to this activity. Have students complete the table but discuss what they notice about their answers and what they wonder.

3.5.2 Refresher: Scatter Plots and Their Data

Component Overview: Students will engage in a refresher on scatter plots and their data, leveraging essential vocabulary and completing a statement using those terms.



3.5.2 Refresher: Scatter Plots and Their Data

Linear equations can also be used to model the relationship between **bivariate data** that shows a linear relationship.



Bivariate data is data that measures two characteristics of a population.

Bivariate numerical data can be represented using a **scatter plot**.



A **scatter plot** is a graph in the coordinate plane representing a set of bivariate numerical data that is used to observe the relationship between two variables.

Each point on a scatter plot represents a specific pair of values from the raw data. The x - and y -values are used to plot each data point in the coordinate plane. When all of the points in a data set are represented, a **line of fit** is often used to model the relationship between the two characteristics in the data set.



A **line of fit** is a line drawn on a scatter plot to estimate the relationship between two sets of data. A line of fit is also known as a trend line.

While different lines of fit can be drawn to model the relationship, the goal is to draw the line of **best** fit to most closely model the relationship and use it to make predictions.



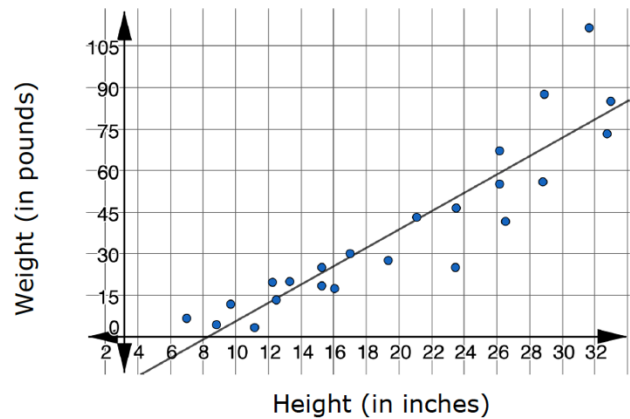
Students learned about scatter plots in a prior course (Grade 7 Accelerated Math or Grade 8 Pre-Algebra). Consider using this refresher as a discussion opportunity. Many students may not be familiar with the term “bivariate”, as they have seen it in a prior grade but likely didn’t use it enough to commit to long term memory. Highlight how this term is important, as it will be used in conjunction with univariate data in later units.



Line of best fit will go deeper in lesson 6. The idea here is that many lines can be fit to data, but only one is the line of best fit.

3.5.2 Refresher: Scatter Plots and Their Data (continued)

- The relationship between the height and weight of 23 dogs is shown.



Complete the statements using the terms in the bank. Some terms will not be used. Do not use a term more than once.

bivariate	decreases	increases
line of best fit	negative	positive
	scatter plot	

The scatter plot shown above represents the relationship between bivariate data. The two variables being compared are the height and weight of 23 dogs. The trend line, or line of best fit, estimates the relationship between these two variables. For this relationship, as dog height increases, the weight of the dog increases, so the trend line has a positive slope.



Consider leveraging a Think-Pair-Share instructional routine with this question. Allow students some individual processing time to try the statement on their own. Then, have them work with a partner to review their answers and come to consensus. Engage the class in a whole-group review of the statements, with correct answers, to clear up any potential vocabulary misconceptions.

3.5.3 Collaboration: Scatter Plots and Patterns of Association

Component Overview: Students will work together to engage in learning more about scatter plots and patterns of association through three, multipart problems.

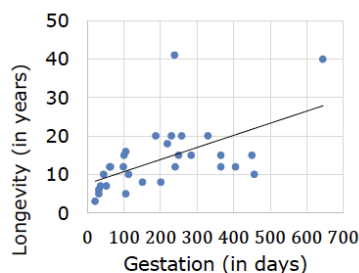


3.5.3 Collaboration: Scatter Plots and Patterns of Association

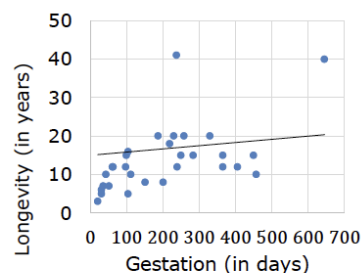
Work with your partner to complete the following.

- The two scatter plots show the same set of data summarizing the average gestation period, in days, and average longevity, in years, for a sample of 30 animals as reported in *The World Almanac and Book of Facts* (McGeveran, 2006). A gestation period is the amount of time for which an animal is pregnant. Longevity refers to how long an animal lives. Each scatter plot shows a different line to model the relationship between gestation period and longevity.

Line of Fit A



Line of Fit B



Discuss with your partner which scatter plot shows a line better fit to model the relationship. Summarize your discussion.

Answers will vary. Line of Fit A is the better fit to model the relationship because it matches the overall pattern of the scatterplot. The Line of Fit B overestimates the longevity of most of the gestation data points.



Notice and Wonder

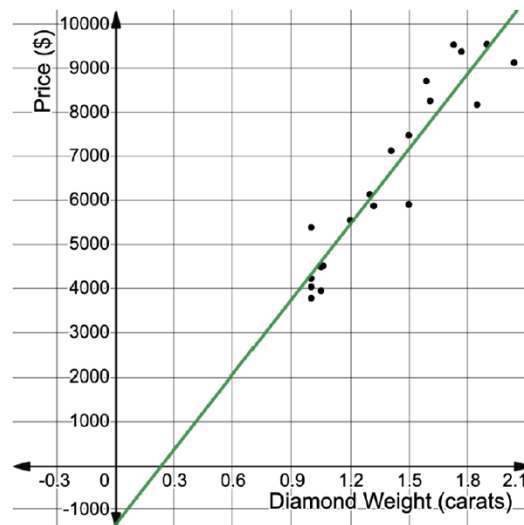
Before students engage with the discussion prompt, consider utilizing a “Notice and Wonder” routine to offer students the opportunity to talk through the context to solidify their understanding of what is being represented on the scatter plots, before discussing the lines of fit.



Students learned to fit lines to data in their prior course. Consider focusing on listening to student discussions for key connections being made. If they are not being made, consider prompting students to dig a little deeper with their analysis.

3.5.3 Collaboration: Scatter Plots and Patterns of Association (continued)

2. The scatter plot shows the relationship between the weight, in carats, and the price, in dollars, of 20 different diamonds.



The equation $y = 5,673x - 1,354$ models the relationship between a diamond's weight and its price based on the data shown. This line of fit can be used to predict the price of a diamond from its weight.

- Define the variables in this context.
 - x represents diamond weight, in carats
 - y represents price, in dollars
- Use the model to predict the price of a diamond that weighs 1.5 carats.
\$7,155.50



To help students understand the importance of defining the variables, consider asking the following questions:

- Why is it important to define the variables in each context?
- How can it help with the analysis and interpretation of the data?

3.5.3 Collaboration: Scatter Plots and Patterns of Association (continued)

- c. Two of the diamonds from the original data set weighed 1.5 carats. The actual price of the two diamonds used to construct the scatter plot are shown in the table.

Weight (carats)	Actual Price (\$)
1.5	7,474
1.5	5,904

Discuss with your partner how the actual prices of these two diamonds compare to the predicted price of a diamond of that weight. Summarize your discussion.

Answers will vary. The first diamond was a little less than the predicted price while the second diamond was substantially less than the predicted price.

The line fit to the data in a scatter plot is used to model the **association** between the variables. It can be used to make estimates or predictions, but the actual values may fall above or below the line.



An **association** is a way to describe the form, direction, or strength of the relationship between the two variables in a bivariate data set. For numerical data, descriptions include linear or nonlinear; positive or negative; strong or weak.



To further analyze the scatter plots, consider asking the following questions:

- Why do you think there would be a difference in the actual outcome and the predicted outcome?
- How are the actual prices represented on the graph?
- How are the predicted prices represented on the graph?



Challenge students to come up with synonyms for “association,” or what comes to mind. This can help students understand its relationship in this context.

3.5.3 Collaboration: Scatter Plots and Patterns of Association (continued)

- d. Complete the statements to describe the patterns of association for the scatter plot.

The relationship between a diamond's weight and price is

☐ linear.
☐ non-linear.

As a diamond's weight

☐ increases,
☐ decreases,

the

price

☐ increases,
☐ decreases,

so the trend is

☐ positive.
☐ negative.

The data

points closely follow the line of fit, so the pattern of

association is

☐ weak.
☐ strong.

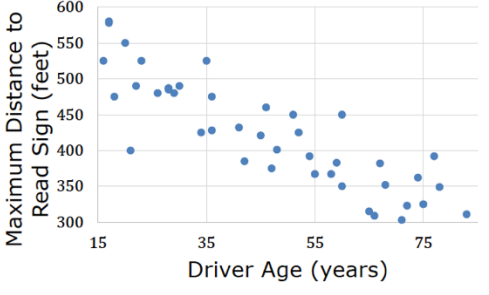
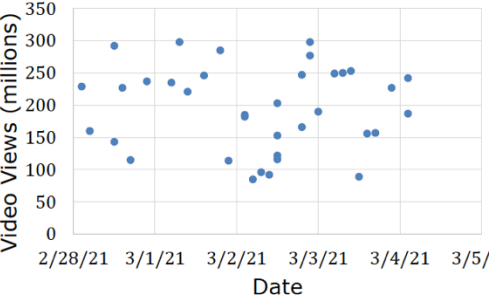


When circulating the room during this activity, look specifically for student responses to question 2 part d. If this is incorrect, they will likely continue to make errors for the remainder of the activity.

3. Complete the table to describe the pattern of association for each scatter plot.

Scatter Plot A	
Pattern of Association	☒ linear ☐ non-linear ☒ positive ☐ negative ☐ neither ☒ strong ☐ moderate ☐ weak

3.5.3 Collaboration: Scatter Plots and Patterns of Association (continued)

Scatter Plot B	
Pattern of Association	☐ linear ☐ non-linear ☐ positive ☐ negative ☐ neither ☐ strong ☐ moderate ☐ weak
Scatter Plot C	
Pattern of Association	☐ linear ☐ non-linear ☐ positive ☐ negative ☐ neither ☐ strong ☐ moderate ☐ weak



Consider taking the information from these three scatter plots and converting them into an anchor chart (or multiple anchor charts) for students to reference as they continue throughout the unit.

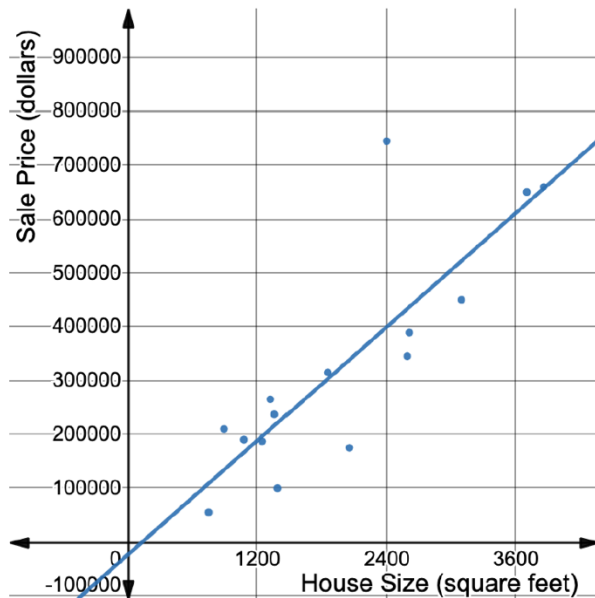
3.5.4 Guided Instruction: Interpreting the Slope and the y -intercept of the Line of Fit

Component Overview: Students will be introduced to interpreting the slope and y -intercept of the line fit to a set of data. For this activity, students will need a partner.



3.5.4 Guided Instruction: Interpreting the Slope and the y -intercept of the Line of Fit

- The scatter plot shows the relationship between the house size, in square feet (ft^2), and the sale price, in dollars, of 16 houses sold in Orlando, FL in 2020.



The equation $y = 176.1x - 23,791$ can be used to model the relationship between size of a house, x , and its sale price, y , in Orlando, FL in 2020.

- Determine the slope and y -intercept of the model.

Slope	176.1
y-intercept	-23,791



Consider starting this activity with a low-entry question of what things they notice about the graph and what things they wonder.



Why is a line of fit needed to determine the slope and y -intercept?

3.5.4 Guided Instruction: Interpreting the Slope and the y-intercept of the Line of Fit (continued)

- b. Discuss with your partner what you think the slope means in this context.

It represents the cost based on the amount of square feet (\$176 times the number of square feet in the home).

- c. Complete the statement.

In this context, the slope means the

- ☐ house size
- ☐ sale price

- ☐ increases
- ☐ decreases

- ☐ \$176.10 per square foot
- ☐ 176.1 square feet per dollar

for

homes in Orlando, FL in 2020.

- d. Trevor and Nadia both considered the meaning of the y-intercept in this context. Each student's thinking is shown in the table.

Trevor's Thinking	Nadia's Thinking
"The y-intercept means the sale price of a house with 0 square feet would have been -\$23,721 in Orlando, FL in 2020."	"The meaning of the y-intercept does not make sense in this context. Houses don't have 0 square feet and they don't sell for negative amounts of money."

Discuss Trevor's and Nadia's thinking with your partner.

- e. Determine an appropriate domain and range for the context, expressed as inequalities.

Domain	$x > 135.099$
Range	$y > 0$



Question 1 part b provides a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight, or misconceptions to address, before moving on to question 1 part c. Use student discussions to determine how much guidance is needed when completing question 1 part c.



Question 1 part d provides an opportunity for students to engage with MA.K12.MTR.4.1, as students will analyze the mathematical thinking of others and discuss their thinking with peers. Listen to students' discussion to determine whether they are making connections between both students' provided thinking, and how they can be used together to make sense of the meaning of the y-intercept in the model.

Discuss these concepts with the whole class before leading into how to determine and express an appropriate domain and range for this context.

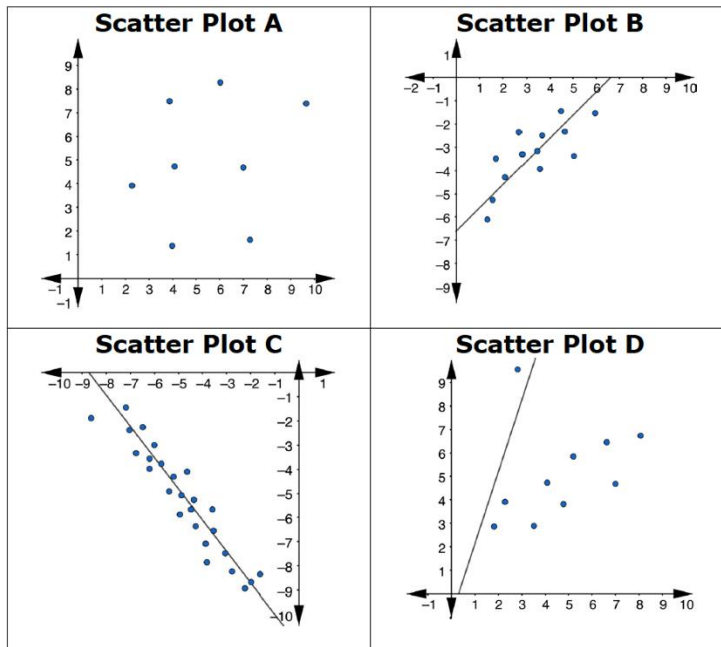
3.5.5 Wrap-Up: Cool Down

Component Overview: Students will consider four scatter plots to determine which one doesn't belong. They will have the opportunity to identify two different scatter plots and make arguments for why either doesn't belong.



3.5.5 Wrap-Up: Cool Down

Four scatter plots are shown.



- Complete each statement to explain why one of the scatter plots doesn't belong with the others. Use a different scatter plot to complete each statement.
 - Scatter plot A doesn't belong because ...
It doesn't include a line of best fit; it is the only one with a weak pattern of association.
 - Scatter plot C doesn't belong because ...
It is the only scatter plot with a negative pattern of association.



Student performance on this activity can help determine what pieces, if any, from the lesson students may still be struggling with and need additional support.

This task could be revisited in future lessons to determine if additional understanding has been solidified as students refine their arguments for which doesn't belong.



Arguments could be made for why any of the scatter plots do not belong with the other. Consider which vocabulary and concepts students use to make their arguments, and what that may mean about which concepts they feel most confident expressing.

Additional correct answers could include:

- Scatter plot A doesn't belong because there is no line of fit drawn; there is a weak pattern of association.
- Scatter plot B doesn't belong because the data shown is all in the fourth quadrant; the y-values are all negative, while the x-values are positive.
- Scatter plot C doesn't belong because it is the only one with a negative pattern of association; the association is strongest for this set of data; the x- and y-values for all data points are negative; all data points in the set are in the third quadrant.
- Scatter plot D doesn't belong because the line drawn does not seem to fit the data; there seems to be one data point that doesn't fit with the others.